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**Re: Letter of Collaboration**

I am pleased to confirm my collaboration with Prof. Maarit Korpi-Lagg and their team, headquartered at the HPC lab in the department of Computer Science at Aalto University, on investigating dust processing in a turbulent interstellar medium (ISM) using the extended multi-scale code-coupling framework.

This project addresses an important tension in astrophysics: the observed dust abundances in starburst galaxies do not agree with current theoretical predictions. The latter are based on the expectation that SN blast waves ought to be very efficient at destroying the dust. Intriguingly, recent results from high resolution but small-scale simulations of the ISM by the present team suggest that the dust destruction rate in the ISM is overestimated and that the actual destruction rate turns out to be significantly lower when one accounts for the turbulence and magnetic fields in the ISM. In these simulations, the magnetic field is the result of small-scale dynamos and the resultant fields are weaker than observed. The expectation is that the weak field generated by small-scale dynamos is almost certainly augmented by magnetic fields on larger scales that are observed in real galaxies. The latter are thought to arise self-consistently via large-scale dynamo effects. Modelling the large-scale magnetic field self-consistently, however, would require simulations with larger domains evolved over gigayears and including the microphysics of dust processing in these larger simulations is, from a computational perspective, prohibitively expensive. Therein lies the challenge: how can one jointly model the small-scale dust microphysics and capture the relevant larger-scale physical processes in a consistent fashion?

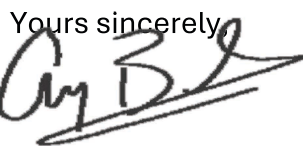
The proposed research programme aims to do this via a multi-scale simulation framework where simulations with successive smaller box sizes (and hence, higher spatial/temporal resolution) essentially zoom in on a small volume within the large-scale simulation. In this framework, the successive smaller volumes inherit large-scale conditions from their parent box. By coupling the simulations in this manner, it is possible to realistically model the physical processes across the full hierarchy of relevant physical and temporal scales and achieve a more complete and physically consistent picture of dust evolution in the interstellar medium.

Apart from leveraging the multi-scale simulation framework, the programme will also leverage advanced machine learning and artificial intelligence methods to enhance the efficiency and the scientific reach of the simulations. These tools will enable automated feature detection, adaptive selection of regions of interest, and intelligent management of computational resources, facilitating real-time data analysis and more effective coupling between simulations across the scales.

The proposed program is timely, high impact, and scientifically very important. JWST's observations are leading to unprecedentedly detailed maps of the 3D structure of the interstellar dust and gas in different galactic environments. The proposed programme has the potential to substantially advance our understanding of how these structures and features form. At the same time, JWST observations are indicating a greater than anticipated abundance of dust-obscured galaxies in the early universe, with many of these being galaxies with high star formation rates. Current models of galaxy evolution cannot account for these observations because – as already noted – of the inherent assumption that SN blast waves ought to be very efficient at destroying the dust, an assumption that may not hold under more realistic conditions.

I am specifically interested in the revised understanding of dust evolution that this programme promises, and its potential impact on our models of galaxy evolution. My research interest (and expertise) is in modeling galaxy evolution using cosmological hydrodynamic numerical simulations and I am convinced that improving the dust modeling in these simulations is key to mitigating the current tension between theory and observations. Moreover, if the dust can survive the SN blast waves and can be successfully expelled into the circumgalactic medium, it can have a significant impact on the latter's cooling profile and that in turn, would further impact galaxy evolution.

To conclude, I would like to, once again, confirm my strong interest in this project and my commitment to contribute my experience and expertise to this initiative. I commit to contributing actively to the scientific discussions of the results as well as their wider implications, and to ensuring that the results are disseminated widely within the astrophysics community

Yours sincerely,  


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